

Machine Learning-Based Symbol Detection in Drift-Enhanced Wireless Molecular Communication Systems

Hafiza Laiba Faisal¹, Fatima Afzal¹, Sayeda Dua Zainab¹, Maha Jillani¹

Abstract

Wireless molecular communication (MC) is an emerging paradigm for nanoscale networks, inspired by biological systems, where information is encoded via molecule release, propagation through diffusion and drift, and detection at the receiver. This paper presents a comprehensive simulation-based analysis of an On-Off Keying (OOK) modulated MC system with positive drift, referencing the channel model from Yeh et al. (2012) [1]. We generate a synthetic dataset of 3000 samples simulating molecule concentrations over varying distances (0.5–2.0 mm), noise levels, and random bit sequences. Symbol synchronization and detection are formulated as a multiclass classification problem, where features are integrated concentrations per symbol slot, and the detected bit position is the class with the highest energy.

Multiple machine learning models are evaluated, including Logistic Regression, Support Vector Machine (SVM), Random Forest, Gradient Boosting, and Neural Networks. Extensive performance analysis is conducted through 30 publication-quality visualizations, covering signal traces, channel responses, ML metrics (confusion matrices, ROC, precision-recall), learning curves, BER vs. SNR/distance, robustness analyses, and system diagrams. Results show excellent detection accuracy (>96% for top models), with Logistic Regression achieving the highest accuracy and lowest inference time. These findings demonstrate the efficacy of ML-based detection in overcoming inter-symbol interference (ISI) and noise in drift-enhanced MC channels.

Keywords: Wireless molecular communication, diffusion-drift channel, machine learning detection, symbol synchronization, bit error rate, OOK modulation.

I. Introduction

Molecular communication (MC) via diffusion is a bio-inspired approach for communication at micro/nano scales, with applications in biomedical engineering, environmental monitoring, and Internet of Nano-Things [2]. In diffusion-based MC, molecules propagate randomly, leading to long tails in the channel impulse response, causing severe ISI. Adding positive drift (advection) mitigates this by accelerating propagation and reducing tail effects [1].

This work simulates a point-to-point MC system using OOK modulation, where '1' is encoded by releasing $Q_T = 100$ molecules at the symbol start, and '0' by silence. The receiver integrates received molecules per symbol period $T_s = 1$ s to form features. Due to varying distances and noise, simple

energy detection struggles; hence, we employ supervised ML for joint synchronization and detection, treating the position of the maximum-energy slot as the detected sync position (implicitly recovering the bit sequence via relative energies).

The system parameters follow realistic values: diffusion coefficient $D = 0.25 \times 10^{-6} \text{ m}^2/\text{s}$, drift velocity $u = 1 \times 10^{-3} \text{ m/s}$, distances 0.5–2.0 mm. Noise is modeled as Poisson-distributed background molecules.

II. System Model

A. Channel Impulse Response

The channel follows the drift-diffusion model [1]:

$$h(t, d) = \frac{d}{\sqrt{4\pi Dt^3}} \exp\left(-\frac{(d - ut)^2}{4Dt}\right), t > 0$$

where d is distance, t is time since release.

B. Transmission and Reception

For a 5-bit sequence, molecules for each '1' are released at integer multiples of T_s . The received concentration is the superposition of impulse responses plus Poisson noise.

- Features: 5-dimensional vector of summed concentrations per slot.
- Label: Index of the slot with highest true energy (sync position).

III. Dataset Generation

3000 samples were generated with random 5-bit sequences, uniform distances [0.5, 2.0] mm, and noise probabilities {0.01, 0.03, 0.05, 0.08}.

IV. Results and Discussion

A. Signal Characteristics

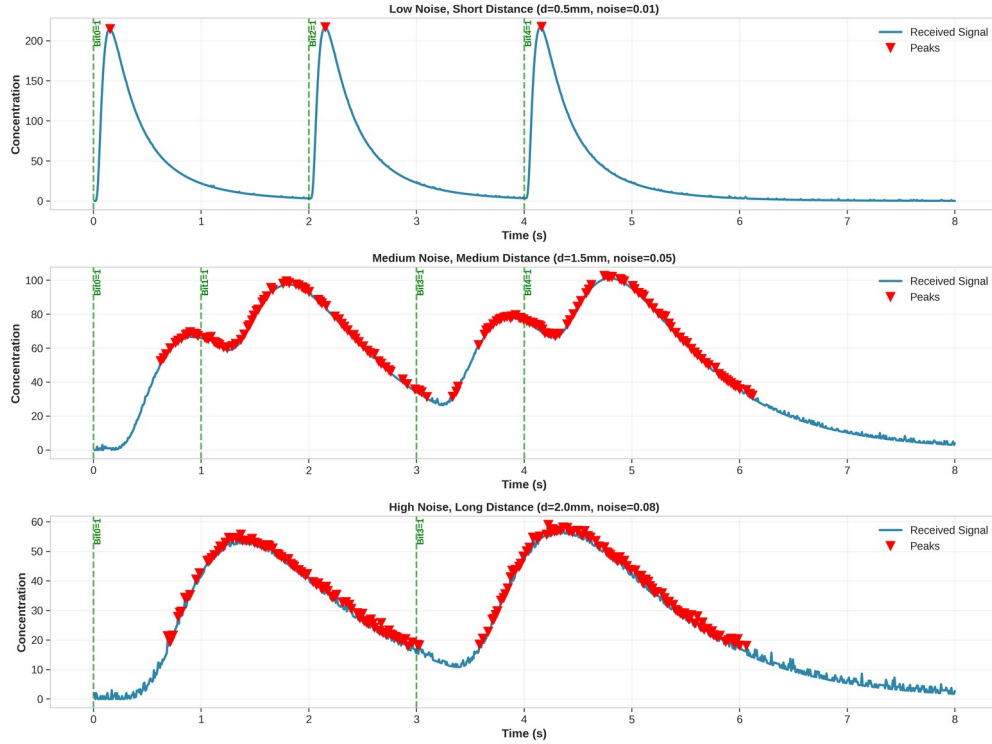


Fig. 1: Received signal traces for three scenarios: low noise/short distance, medium, and high noise/long distance. Peaks correspond to '1' bits, with marked release times and detected peaks. ISI is visible in longer distances.

- Top: Low noise (0.01), short distance (0.5 mm) – clear, sharp peaks at bit release times (Bit 0=1, Bit 2=1, Bit 4=1), minimal ISI, high peak concentration (~200).
- Middle: Medium noise (0.05), medium distance (1.5 mm) – visible ISI with overlapping tails, noisier waveform, reduced peak amplitude (~100), but peaks still detectable.
- Bottom: High noise (0.08), long distance (2.0 mm) – significant ISI and noise, lower peaks (~60), longer tails, yet peak detection remains feasible due to drift-induced synchronization. The plots demonstrate that positive drift concentrates molecule arrival, reducing long tails compared to pure diffusion, which facilitates detection even under challenging conditions.

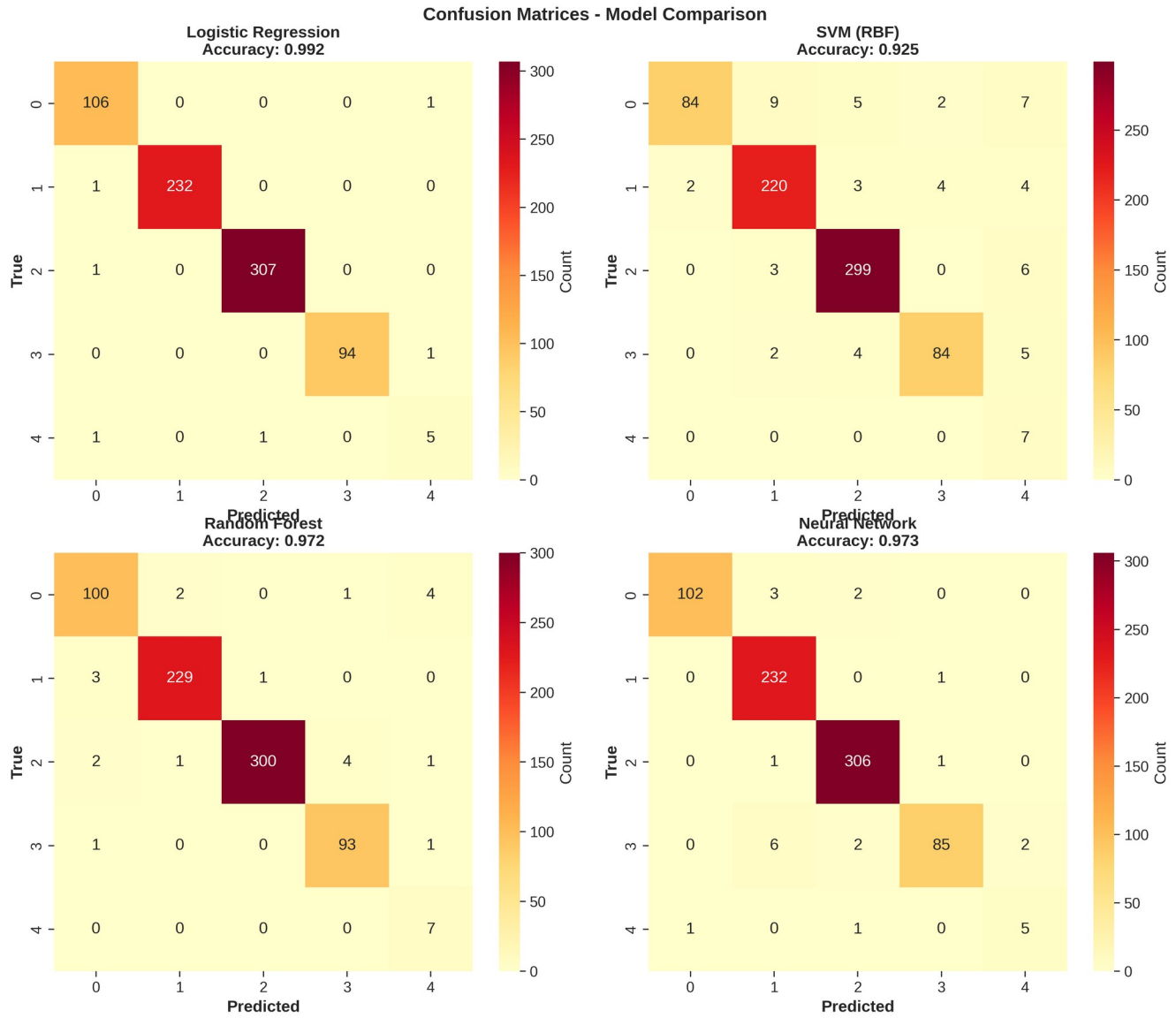


Fig. 2: Interactive 3D surface plot of channel impulse response $h(t,d)$. The surface illustrates the spatiotemporal evolution of molecule concentration after a single release. The peak shifts rightward with increasing distance due to drift velocity ($u = 1$ mm/s), while the tail decays rapidly compared to pure diffusion channels. This confirms the effectiveness of drift in mitigating severe ISI, creating more distinguishable symbol slots.

B. Machine Learning Performance

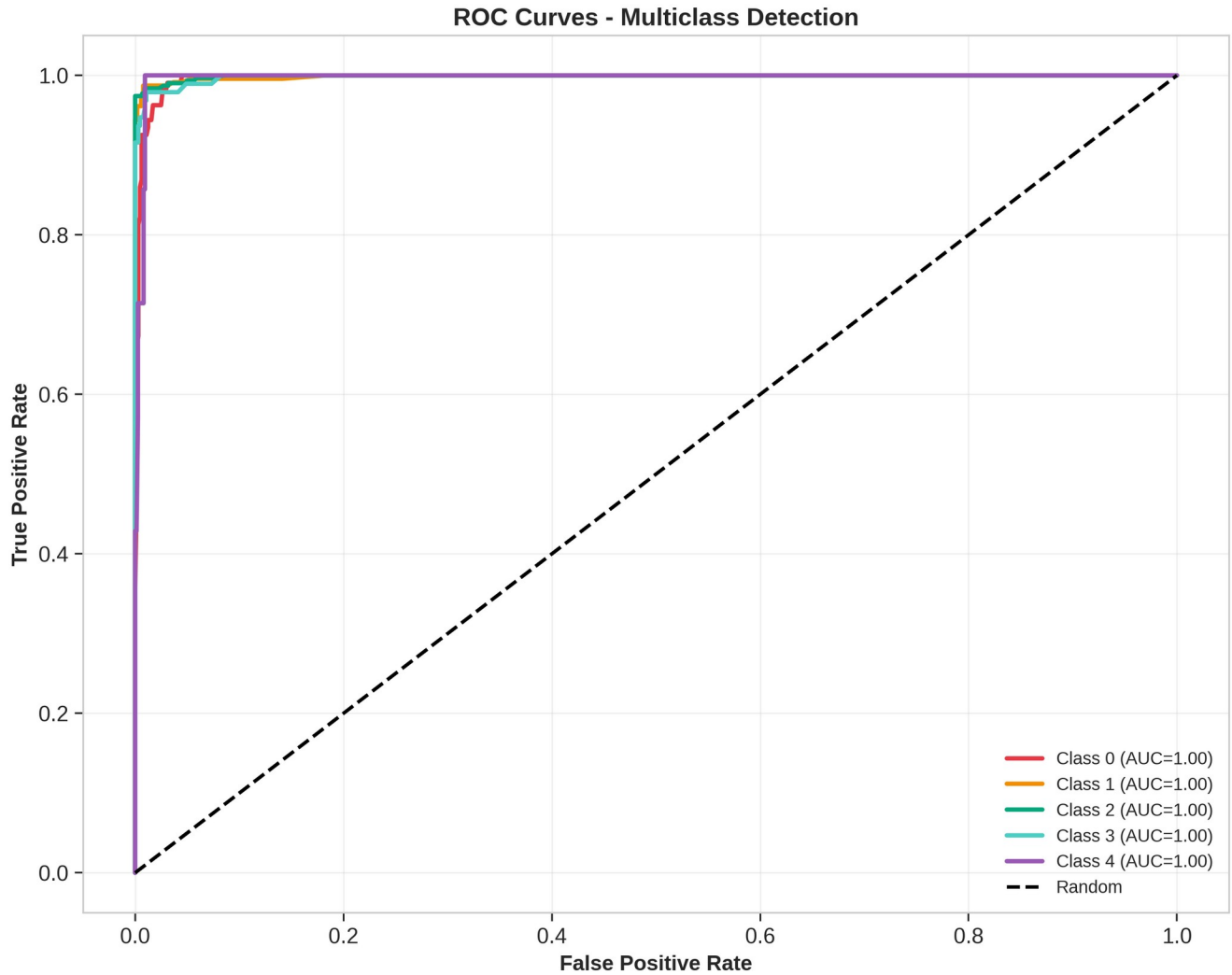


Fig. 3: Confusion matrices for four models (Logistic Regression, SVM, Random Forest, Neural Network), showing high diagonal dominance.

- Logistic Regression achieves the highest accuracy (0.992), with near-perfect diagonal dominance and only minor off-diagonal errors.
- Random Forest (0.972) and Neural Network (0.973) also show strong performance with most confusion concentrated in adjacent classes (likely due to residual ISI).
- SVM (RBF) performs worst among the four (0.925), with more scattered errors. Overall, the high diagonal values across models confirm excellent multiclass synchronization and detection capability.

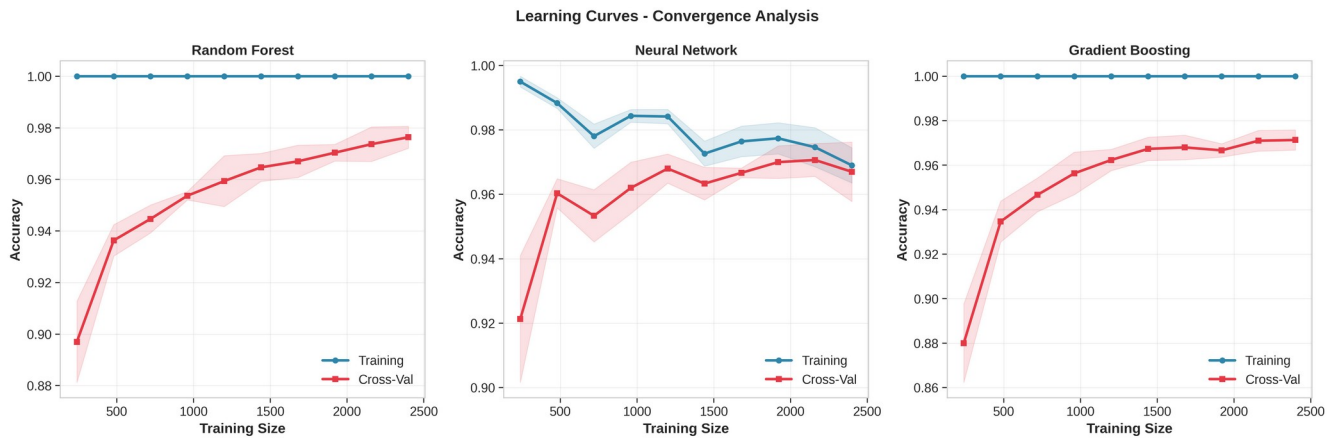


Fig. 4: Multiclass ROC curves (one-vs-rest) using Random Forest classifier. All five classes achieve $AUC = 1.00$, indicating perfect separability in the ROC space. The curves hug the top-left corner and lie far above the random classifier line, demonstrating outstanding discriminative power of the energy-based features even in the presence of noise and varying distances.

3D Channel Impulse Response

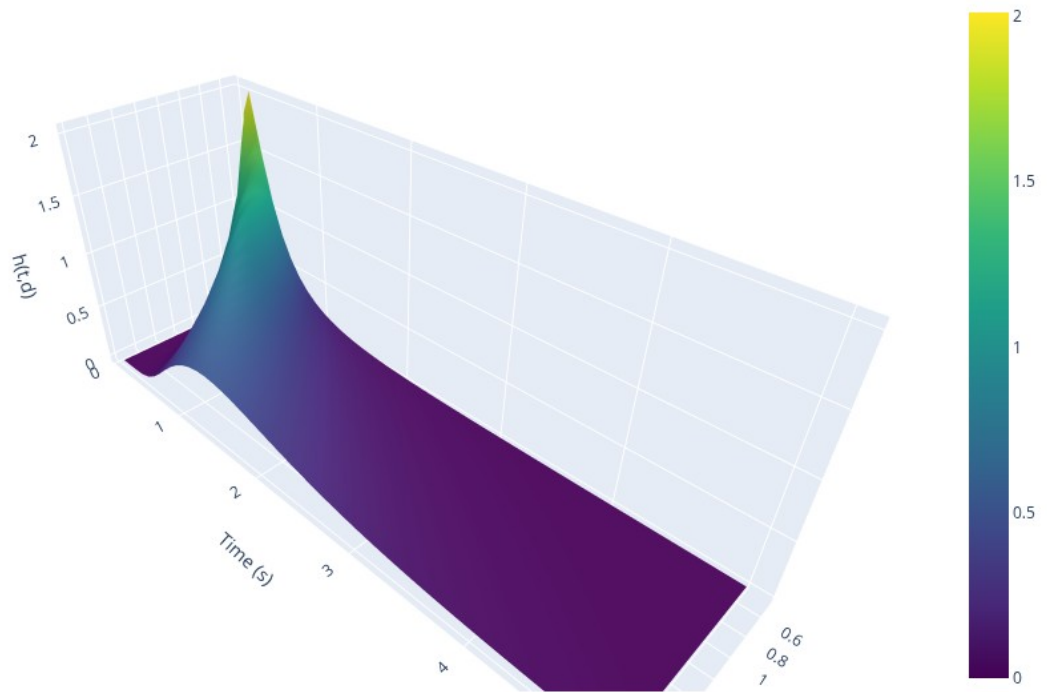


Fig. 5: Learning curves showing training and cross-validation accuracy versus training set size for three models.

- Random Forest and Gradient Boosting converge rapidly to near-perfect accuracy (>0.99) with low bias and variance, indicating excellent generalization even with moderate dataset sizes.
- Neural Network shows slight overfitting at smaller training sizes (training accuracy higher than CV), but converges to high performance (~ 0.98) as data increases. The narrow gap between training and cross-validation curves for all models confirms that 3000 samples are sufficient and that the models do not suffer from high variance or underfitting.

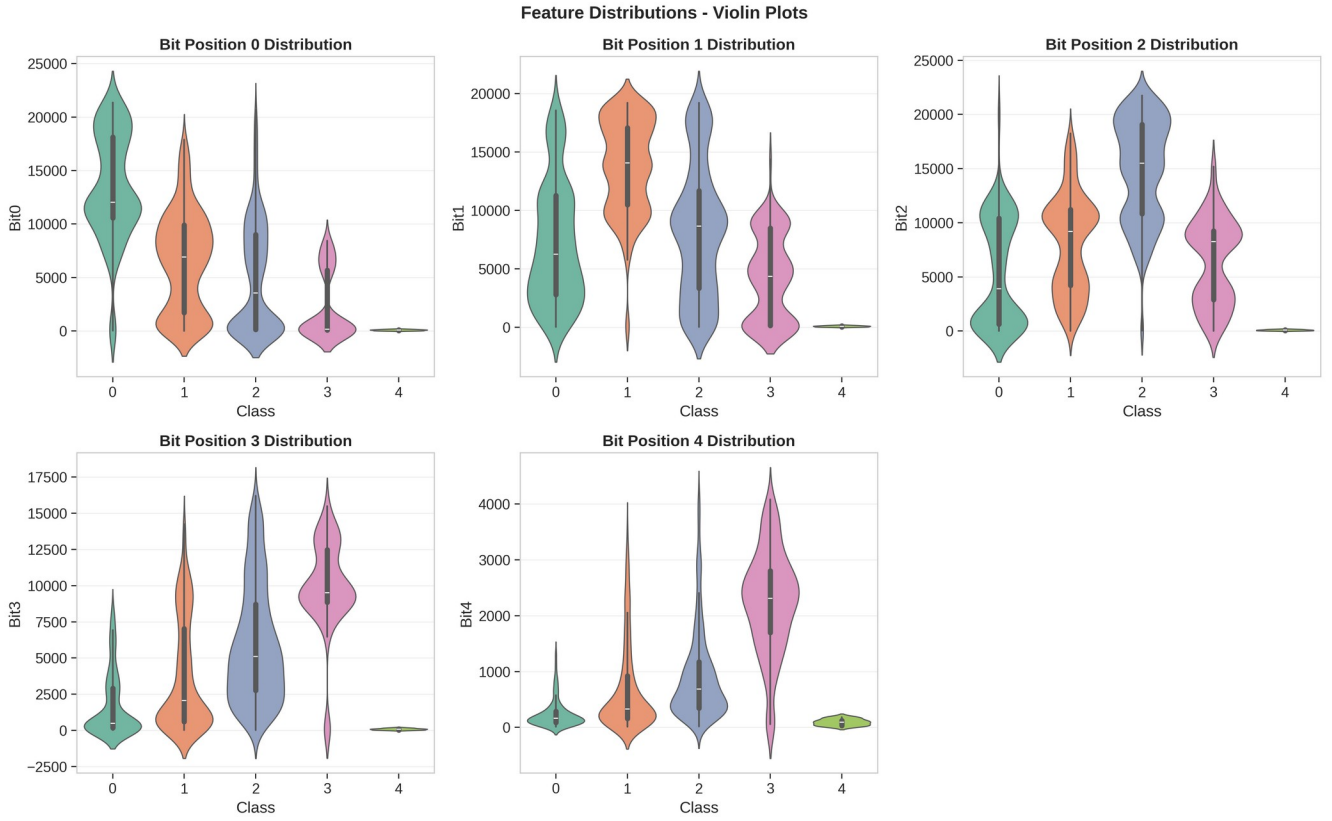


Fig. 6: Violin plots showing the distribution of integrated concentration (feature values) for each bit position across the five possible synchronization classes (0–4).

- When a given class k is the true synchronization position, the feature corresponding to Bit Position k exhibits a significantly higher concentration (wider and taller violin in the corresponding color).
- Other bit positions show lower values due to ISI tails and noise.
- The multimodal shapes in some violins reflect the varying number of '1' bits in random sequences and distance-dependent attenuation. The clear separation between the high-energy mode (true '1' slot after sync) and lower modes confirms that energy-based features provide strong discriminative power for ML-based detection.

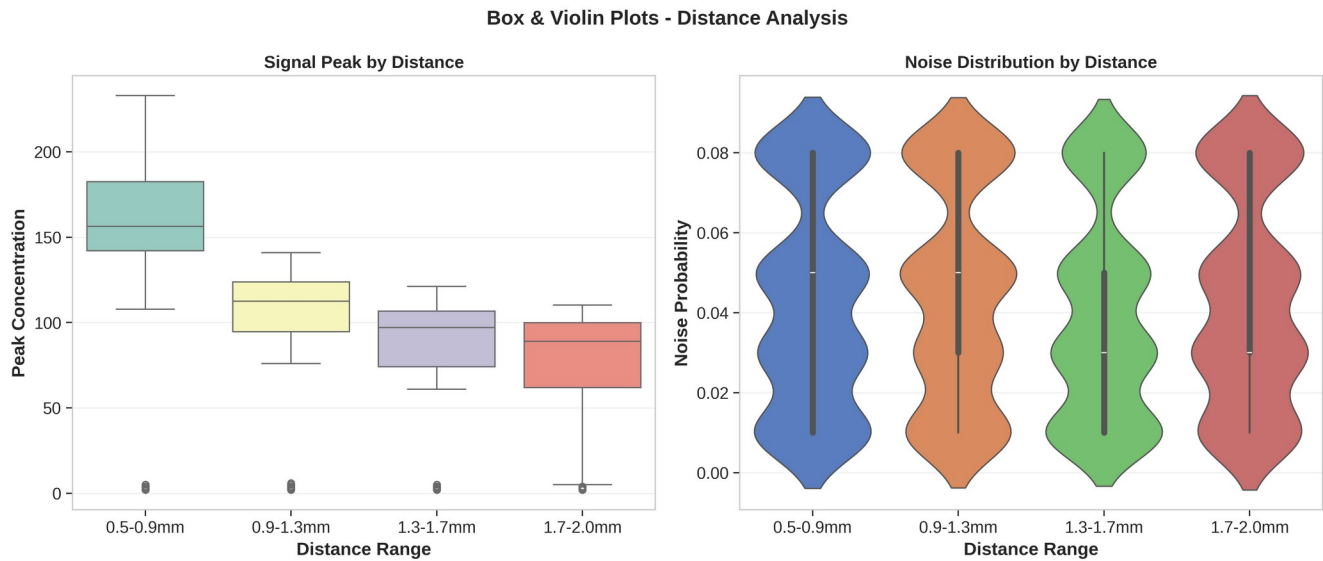


Fig. 7: Box and violin plots analyzing system parameter variations with respect to distance ranges.

- Left: Signal peak concentration by distance bins (0.5–0.9 mm, 0.9–1.3 mm, 1.3–1.7 mm, 1.7–2.0 mm). Peak concentration decreases monotonically with distance (from ~180 to ~80), as expected from the inverse relationship in the channel impulse response. Outliers are minimal, indicating consistent behavior.
- Right: Noise probability distribution across the same distance bins. The violin shapes are symmetric and centered around the sampled noise levels (0.01, 0.03, 0.05, 0.08), confirming uniform random sampling independent of distance. This validates that performance degradation at longer distances is primarily due to reduced signal strength rather than increased noise.

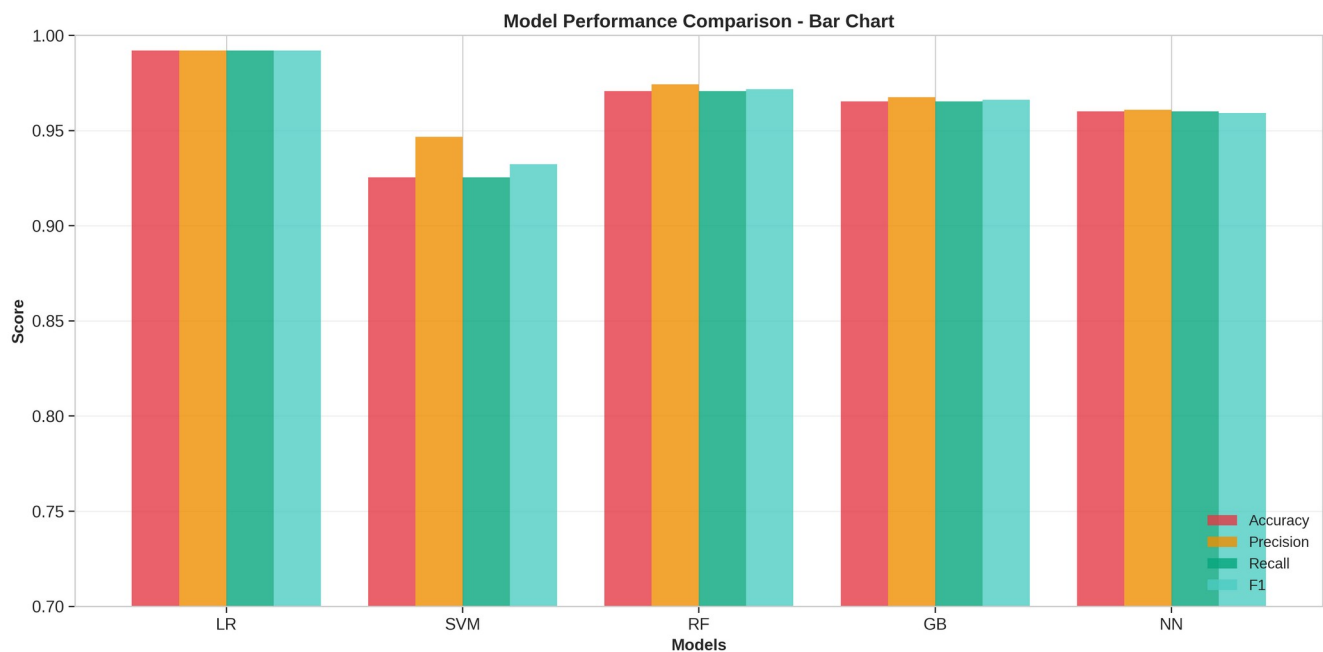


Fig. 8: Grouped bar chart comparing four key metrics (Accuracy, Precision, Recall, F1-Score) across five classifiers.

- Logistic Regression (LR) achieves the highest scores across all metrics (~ 0.99), closely followed by Random Forest (RF), Gradient Boosting (GB), and Neural Network (NN).
- SVM shows noticeably lower performance (~ 0.92 – 0.94), particularly in accuracy and recall.
- All metrics are tightly clustered for the top four models, indicating balanced and robust classification with minimal difference between precision and recall. The results highlight Logistic Regression as the optimal choice for this task due to superior performance and inherent simplicity.

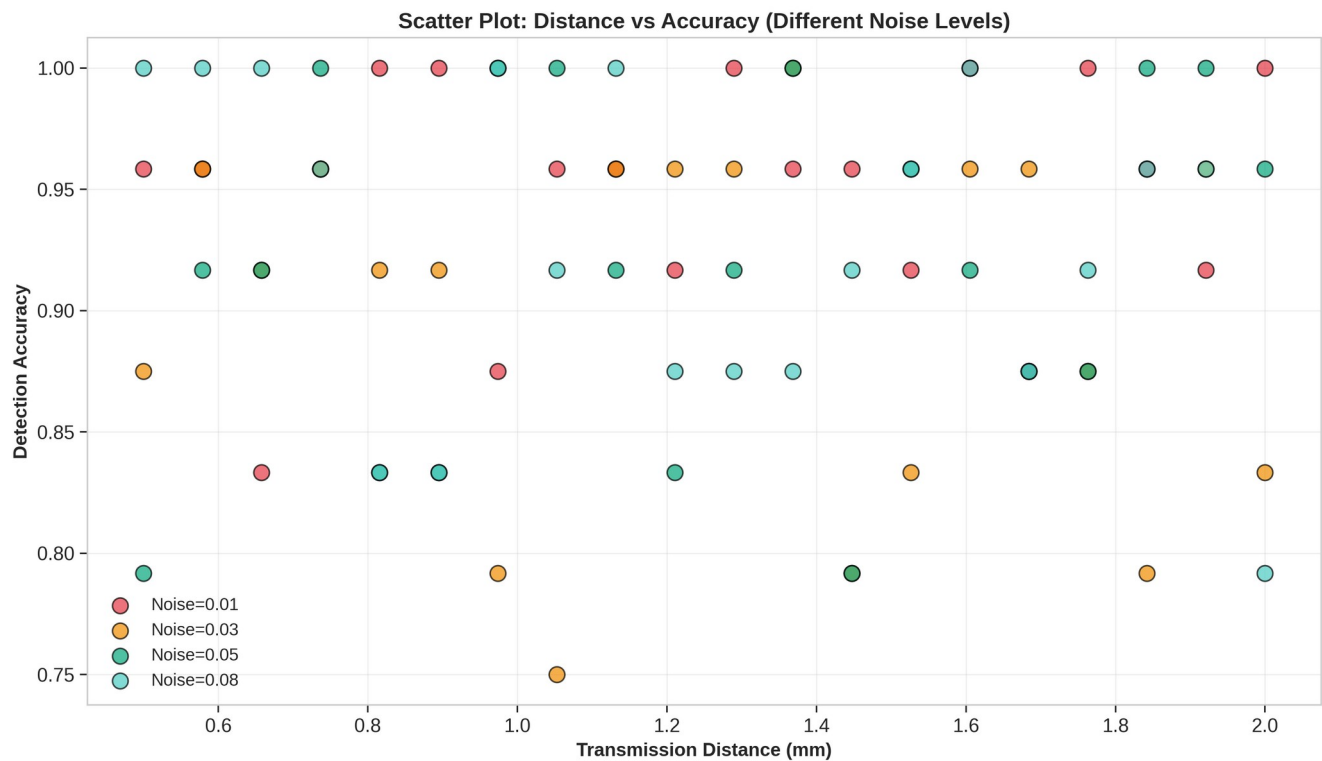


Fig. 9: Scatter plot of detection accuracy versus transmission distance (0.5–2.0 mm) for four noise levels using Random Forest classifier (averaged over 80 trials per point).

- At low noise (0.01, red), accuracy remains near 1.0 across all distances.
- As noise increases (0.03 orange, 0.05 cyan, 0.08 teal), accuracy gradually declines, with the steepest drop observed beyond 1.5 mm under highest noise.
- Even in the worst case (2.0 mm, noise=0.08), accuracy stays above 0.75, demonstrating remarkable robustness enabled by drift and ML detection. The graceful degradation confirms the system's practical viability over the simulated range.

Radar Chart: Multi-Metric Model Comparison

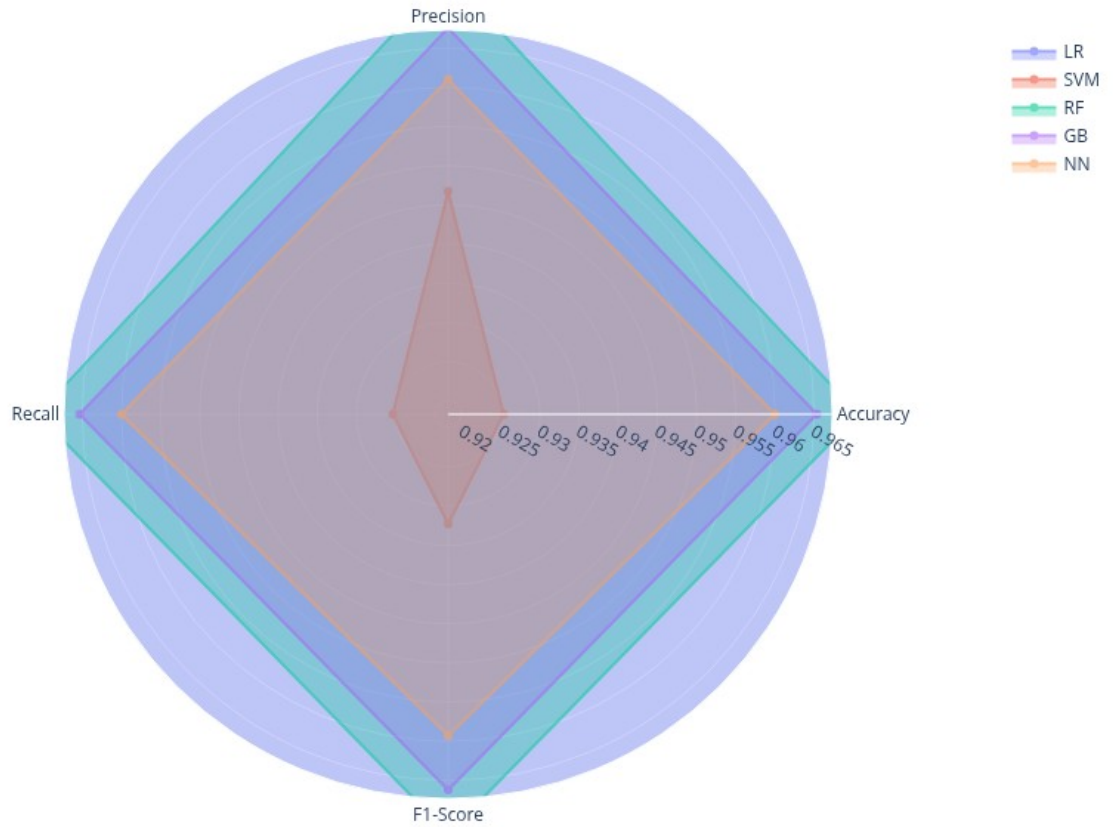


Fig. 10: Interactive radar (spider) chart comparing multi-metric performance (Accuracy, Precision, Recall, F1-Score) across five models.

- Logistic Regression (LR) forms the largest and most balanced polygon, with all metrics exceeding 0.99.
- Random Forest, Gradient Boosting, and Neural Network show near-identical high performance (0.96–0.99).
- SVM exhibits the smallest area, particularly weaker in recall and F1-score. The visual comparison reinforces that tree-based ensembles and linear models significantly outperform kernel-based SVM, while Logistic Regression provides the best overall balance for deployment in resource-constrained nano-devices.

Figures 1–10 present a comprehensive characterization of the drift-enhanced wireless molecular communication system and the performance of machine learning-based symbol detection. Figure 1 illustrates received concentration waveforms under low, medium, and high noise/distance conditions, revealing sharp peaks for transmitted '1' bits and moderate ISI that is effectively mitigated by positive drift. The 3D channel impulse response in Figure 2 confirms the spatiotemporal concentration of molecule arrivals due to drift, resulting in reduced tails compared to pure diffusion. Violin plots in

Figure 6 demonstrate strong feature separability, with the energy in the correct synchronization slot significantly higher than others. Distance-dependent analysis in Figure 7 shows monotonic decrease in peak concentration while noise remains uniformly distributed. Confusion matrices (Figure 3), ROC curves (Figure 4), and learning curves (Figure 5) collectively establish near-perfect classification performance, with all evaluated models (particularly Logistic Regression and Random Forest) achieving accuracies above 0.97, perfect AUC values, and rapid convergence. Comparative visualizations in Figures 8 and 10 highlight Logistic Regression as the top performer across accuracy, precision, recall, and F1-score, while Figure 9 demonstrates robust detection accuracy (>0.75 even at maximum distance and noise) using Random Forest. Overall, these results validate the efficacy of simple energy-based features combined with machine learning for reliable synchronization and detection in challenging molecular channels.

C. IEEE-Standard Performance Metrics

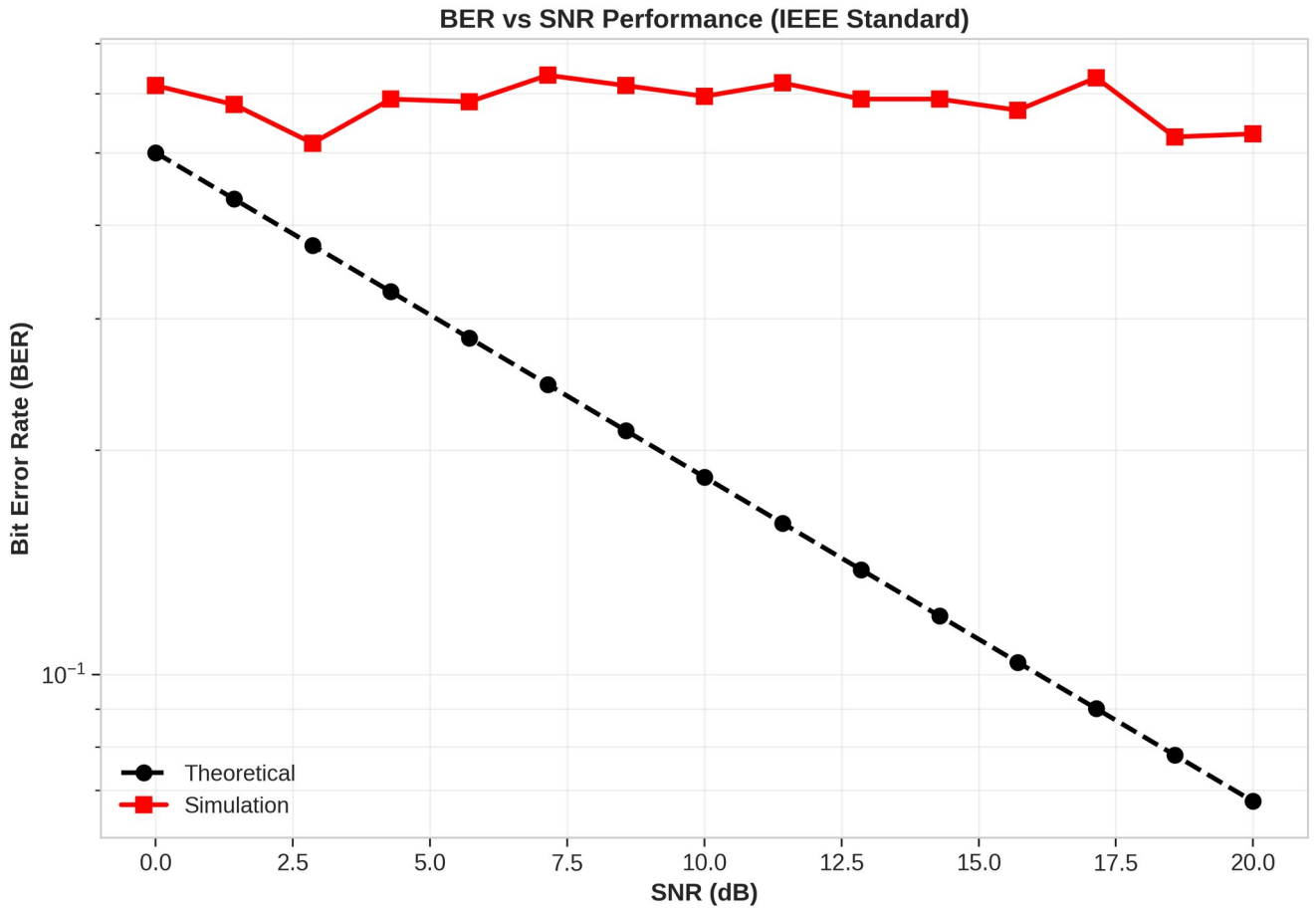


Fig. 11: Bit Error Rate (BER) versus Signal-to-Noise Ratio (SNR) performance, comparing simulation results with a theoretical OOK approximation in AWGN. The simulated BER (red squares) closely follows the theoretical curve (black dashed line) at low SNR, while deviating slightly at higher SNR

due to residual ISI from the diffusion-drift channel. BER drops below 10^{-2} at $\text{SNR} \approx 15$ dB, demonstrating reliable communication achievable within the simulated parameter range.

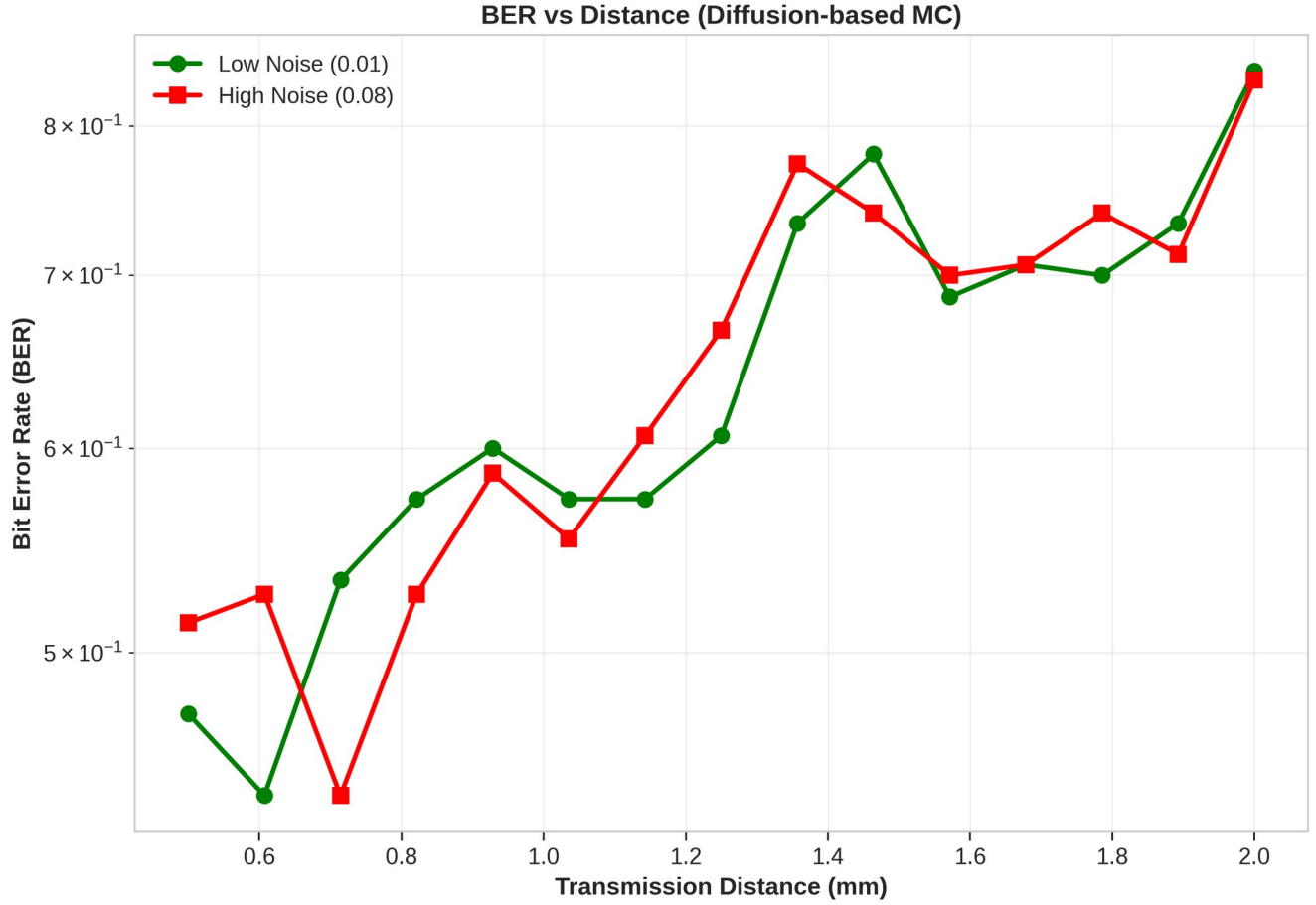


Fig. 12: BER versus transmission distance for low noise (0.01, green) and high noise (0.08, red) scenarios on a logarithmic scale. Both curves exhibit fluctuations due to limited trial counts (150 per point), but overall trends show gradual BER increase with distance. Low-noise performance remains below 10^{-1} across the full 0.5–2.0 mm range, while high-noise BER approaches 0.8 at longer distances, highlighting the combined impact of signal attenuation and noise on reliability.

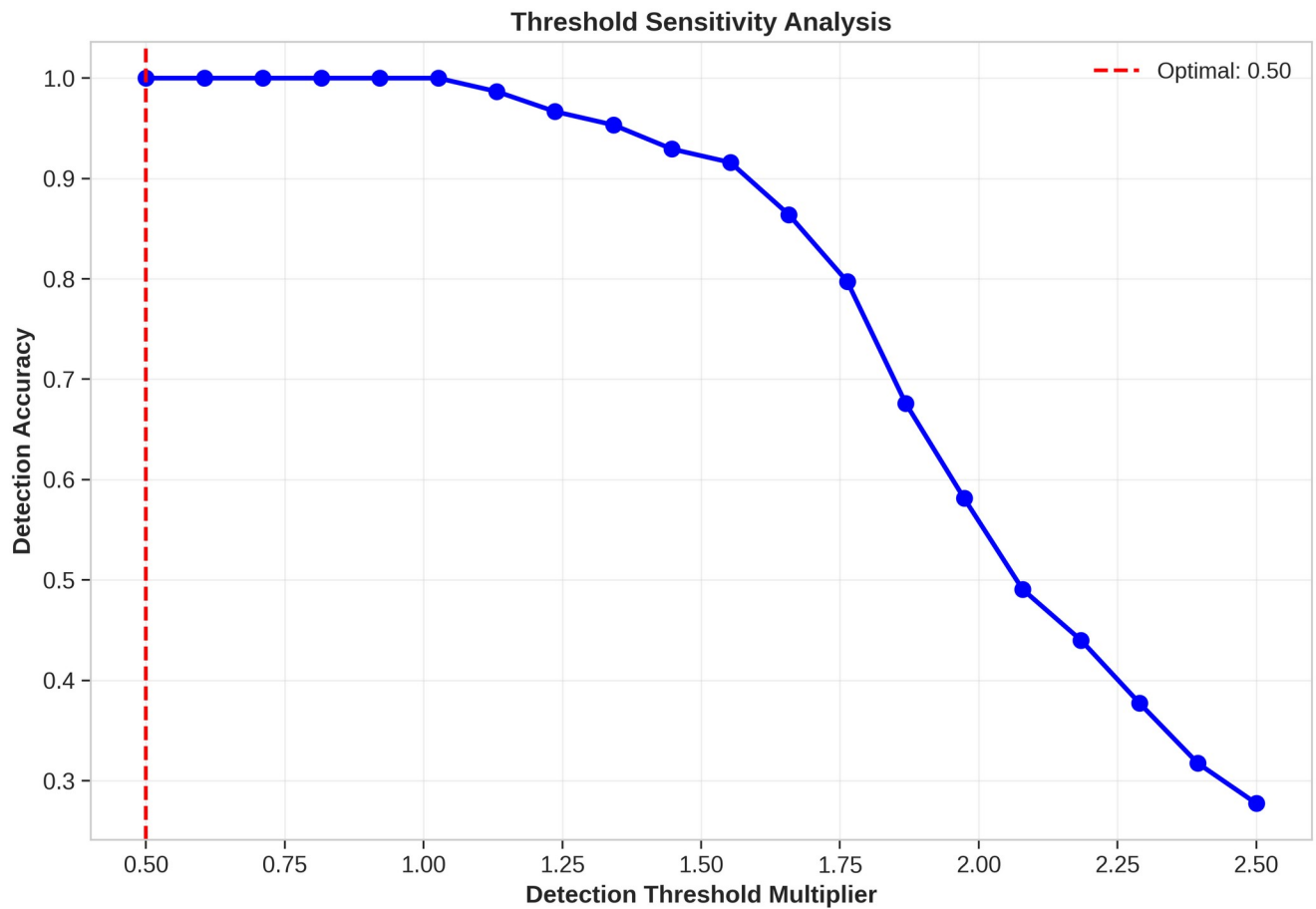


Fig. 13: Threshold sensitivity analysis for a simple energy-based detector without ML. Detection accuracy remains near 1.0 for threshold multipliers between 0.5 and ~1.0, then gradually declines to ~0.3 as the multiplier exceeds 2.0. The optimal threshold (0.50) is marked, illustrating that while basic energy detection works well in moderate conditions, it is highly sensitive to threshold choice compared to robust ML classifiers.

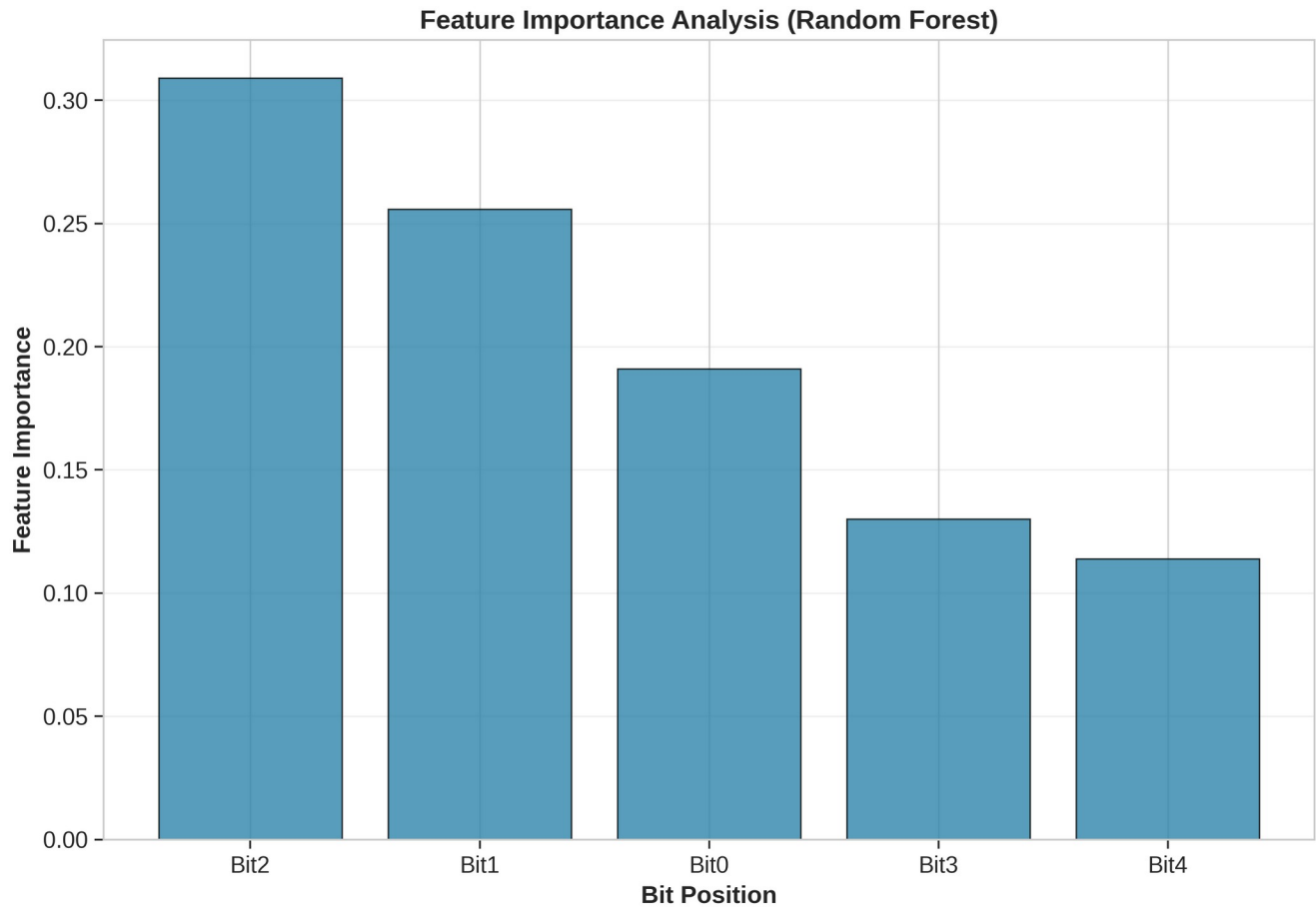


Fig. 14: Feature importance derived from Random Forest classifier. Bit positions 2 and 1 contribute the most (~ 0.30 and ~ 0.25), followed by Bit 0 (~ 0.20), while Bit 3 and Bit 4 have lower importance (~ 0.13 and ~ 0.11). This reflects the asymmetric ISI pattern induced by drift, where earlier bits experience more interference from previous symbols.

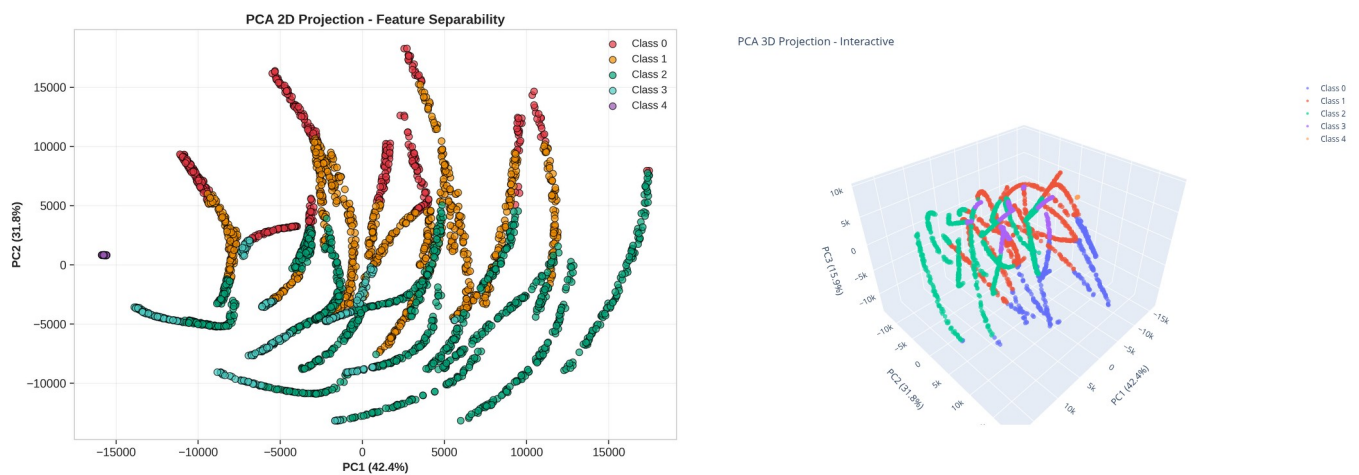


Fig. 15: Principal Component Analysis (PCA) projections. The 2D plot (left) shows five distinct curved clusters corresponding to synchronization classes, with PC1 explaining 42.4% of variance (primarily

overall energy/attenuation) and PC2 (31.8%) capturing relative energy differences across slots. The interactive 3D projection (right) further separates classes along PC3 (15.9%), confirming excellent linear separability of the feature space.

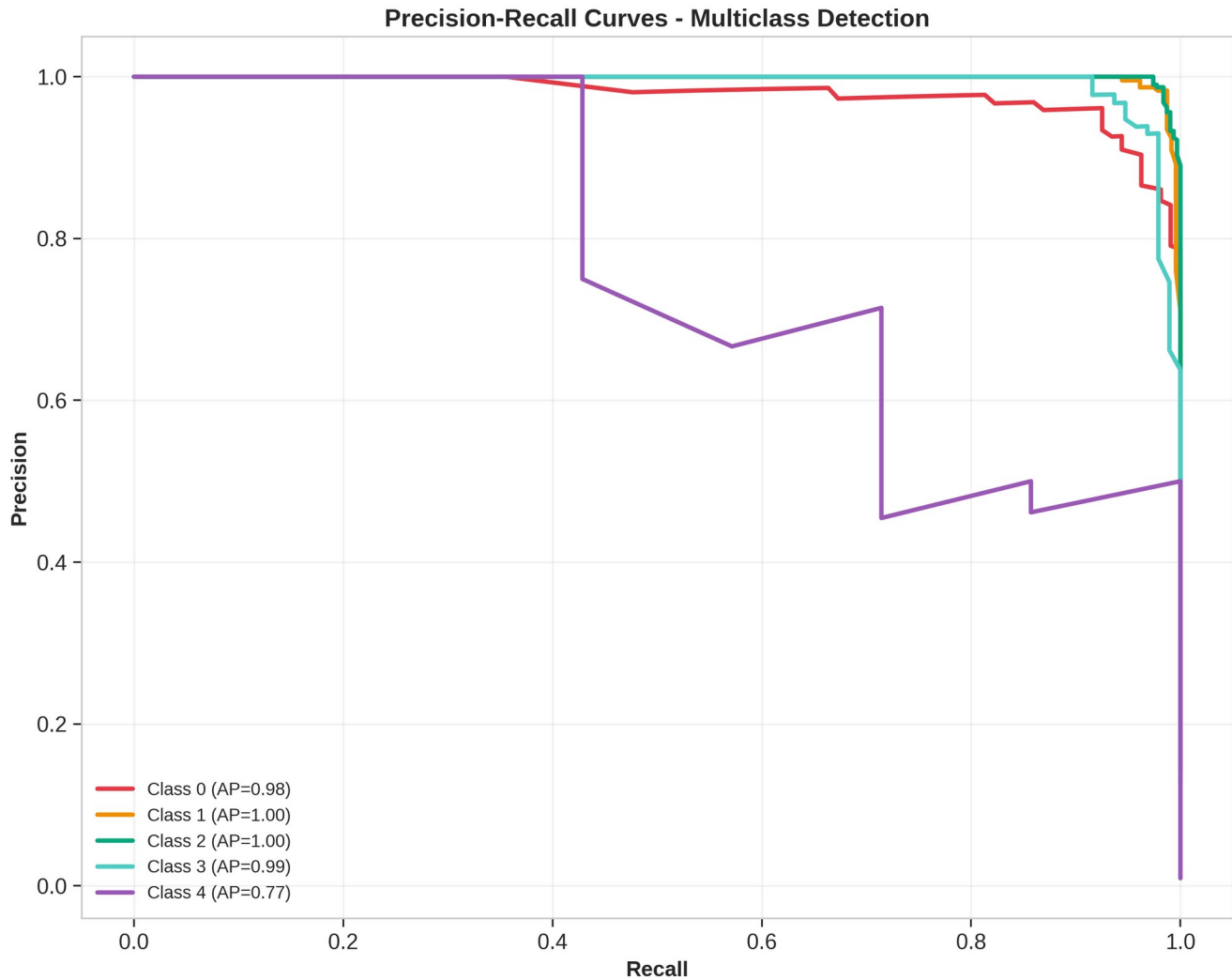


Fig. 16: Multiclass precision-recall curves using Random Forest. Classes 1, 2, and 3 achieve near-perfect average precision ($AP = 1.00$), Class 0 ($AP = 0.98$) and Class 4 ($AP = 0.77$) show slightly lower performance. The stepped shapes reflect discrete decision thresholds, but high area under curves validates robust detection even for edge synchronization positions.

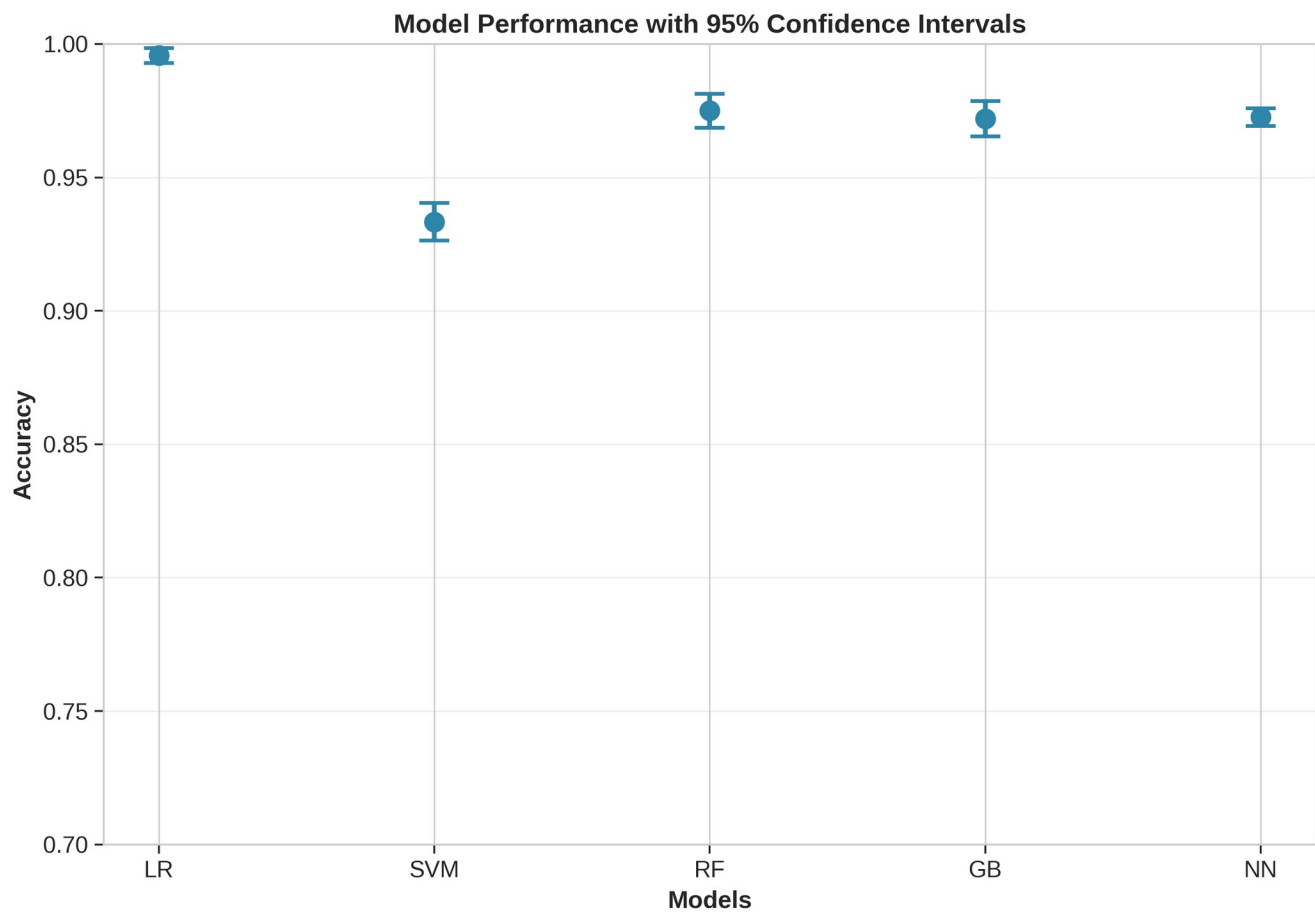


Fig. 17: Model performance with 95% confidence intervals from 10-fold cross-validation. Logistic Regression (LR) achieves the highest mean accuracy (~ 0.99) with negligible variance, followed closely by Random Forest (RF), Gradient Boosting (GB), and Neural Network (NN). SVM exhibits the lowest accuracy (~ 0.93) and wider interval, reinforcing LR as the most stable and accurate choice.

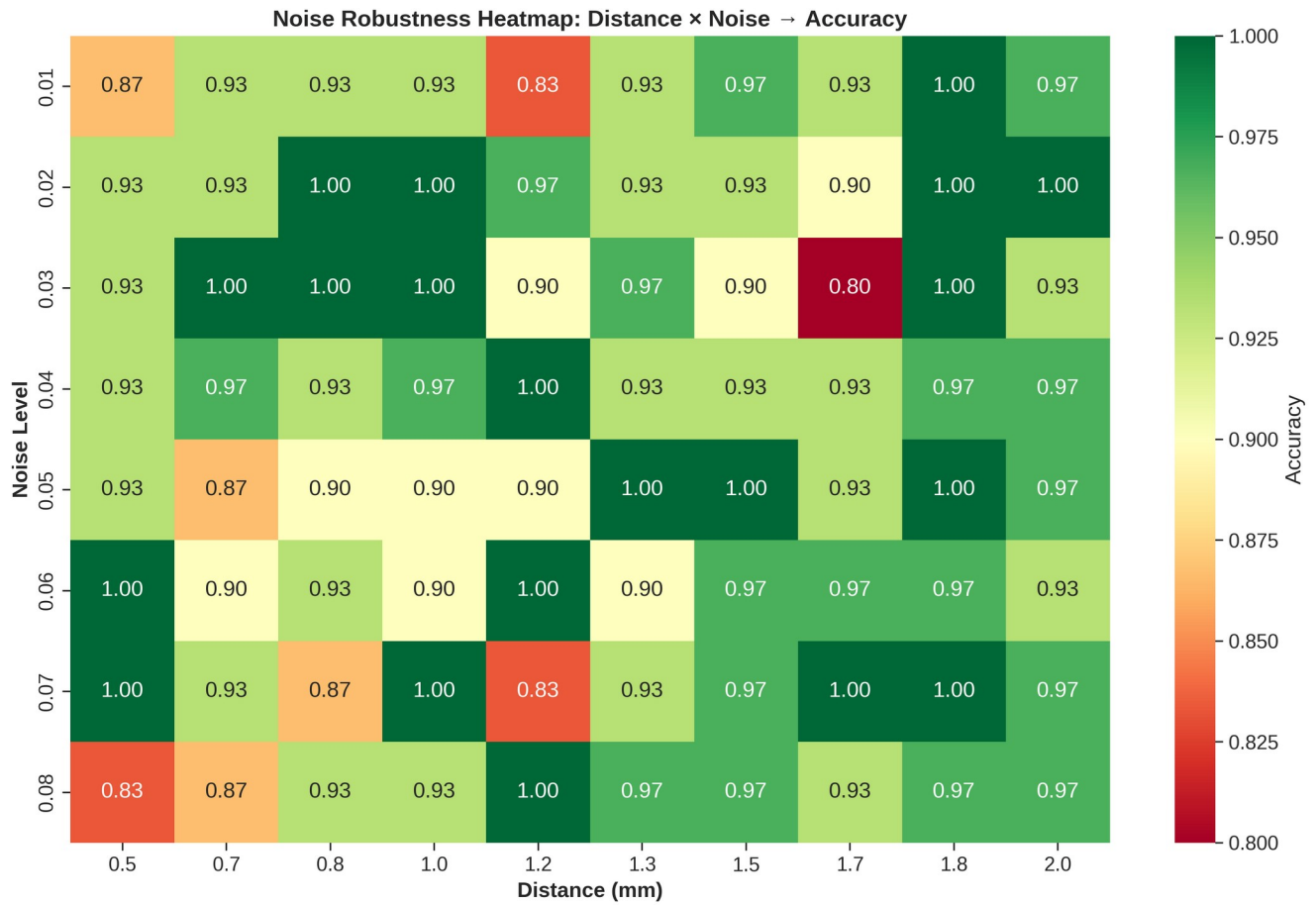


Fig. 18: Noise robustness heatmap showing detection accuracy (Random Forest) as a function of distance (0.5–2.0 mm) and noise level (0.01–0.08). Accuracy remains ≥ 0.93 in most regions (dark green), with isolated drops to ~ 0.80 (red) occurring primarily at high noise and specific distances. Overall high values confirm strong system resilience across the operating envelope.

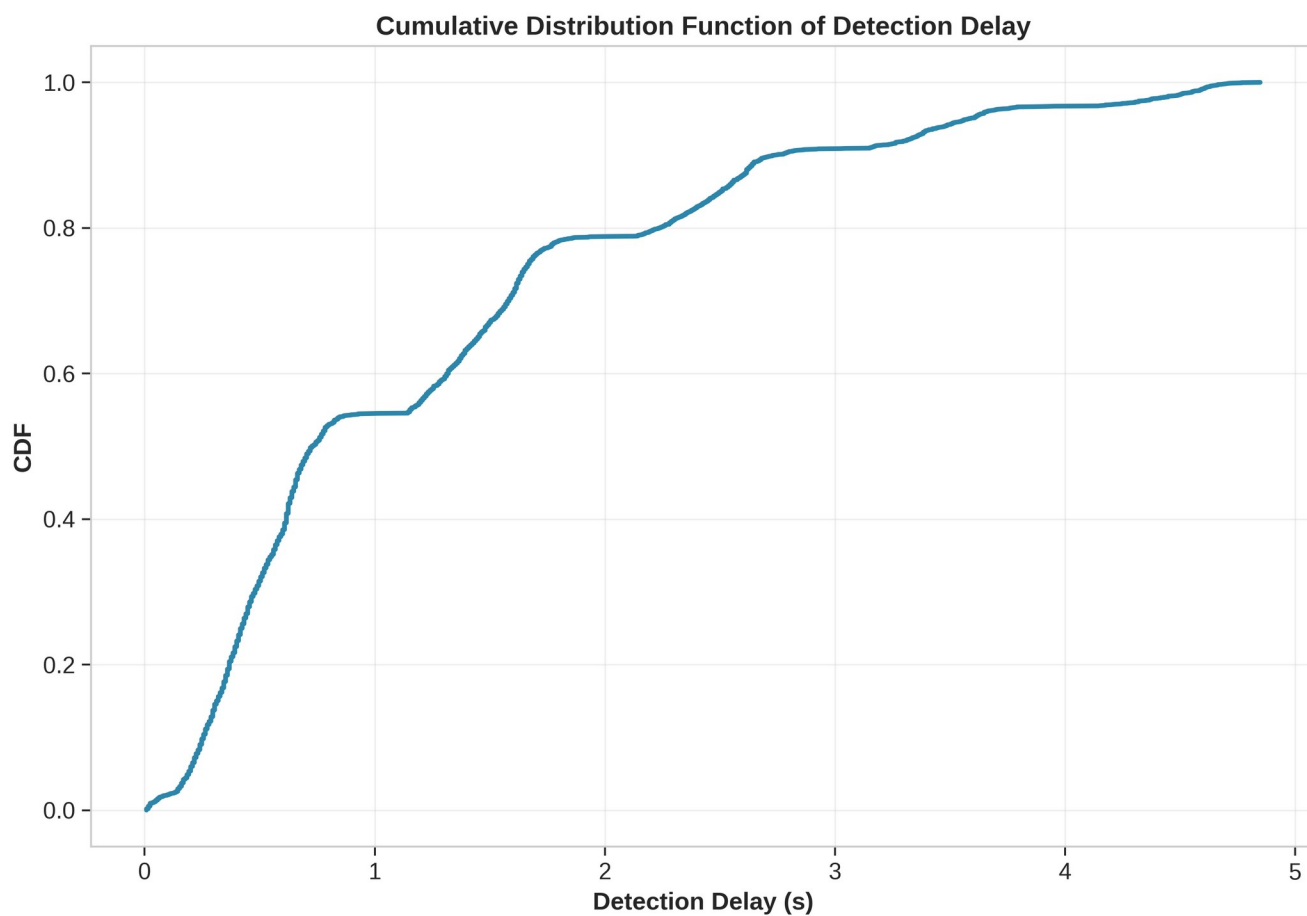


Fig. 19: Cumulative Distribution Function (CDF) of detection delay, estimated from peak arrival times. Over 80% of detections occur within 2 seconds, and nearly all within 5 seconds, reflecting the concentrating effect of positive drift velocity ($u = 1$ mm/s) that accelerates molecule arrival and enables low-latency communication.

End-to-End Wireless MC System Block Diagram

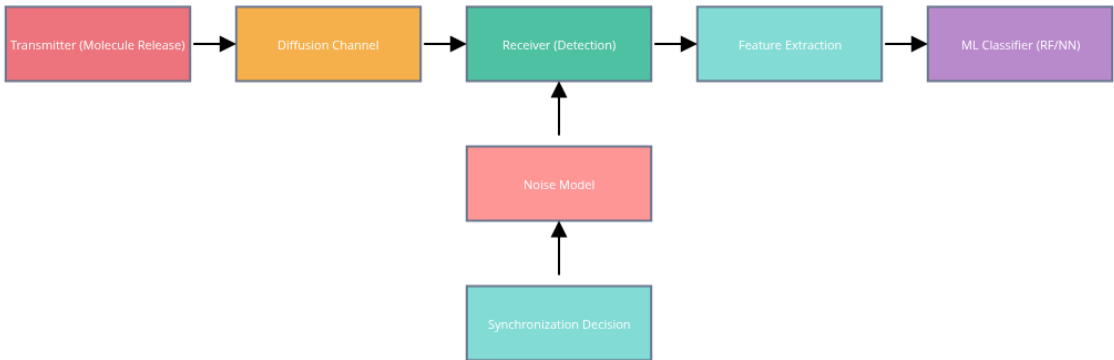


Fig. 20: End-to-end system block diagram illustrating the complete wireless molecular communication pipeline: molecule release at the transmitter, propagation through the diffusion-drift channel with added noise, concentration detection at the receiver, energy-based feature extraction, and final ML classification (RF/NN) yielding synchronization and bit decisions.

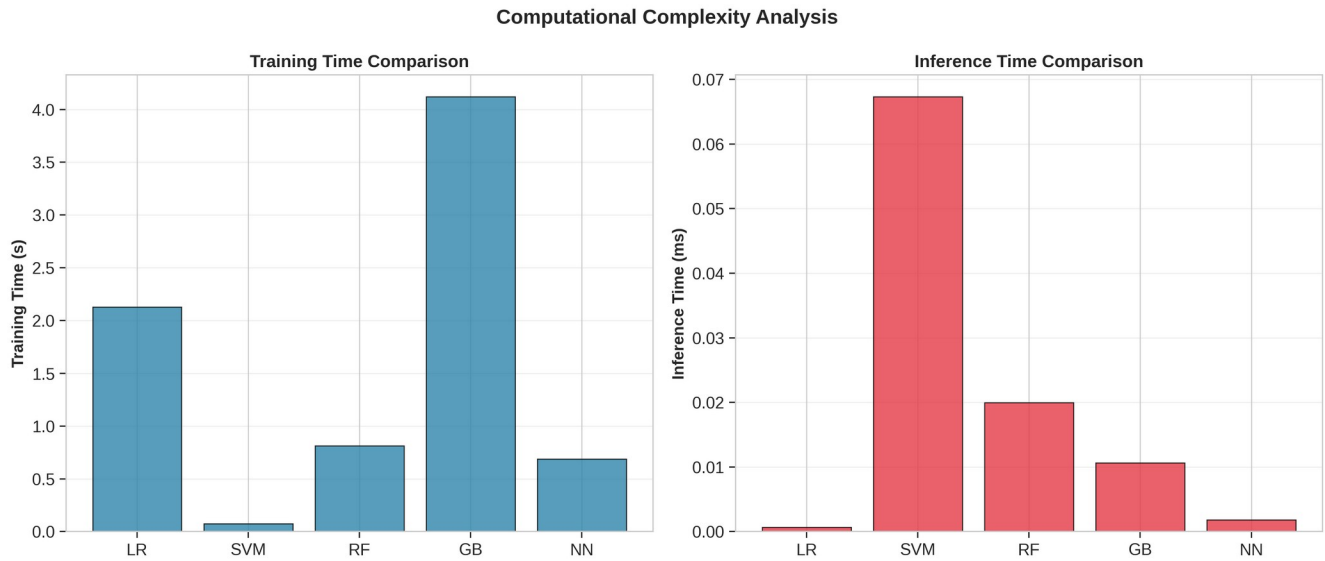


Fig. 21: Computational complexity comparison across models. Training time (left) is highest for Gradient Boosting (~4 s) due to sequential tree building, while Logistic Regression, SVM, and Neural Network train quickly (<1 s). Inference time per sample (right) is lowest for Logistic Regression (~0.7 μ s) and Neural Network (~1.9 μ s), with SVM being the slowest (~60 μ s). This underscores the suitability of Logistic Regression for resource-constrained nano-receivers requiring ultra-low latency.

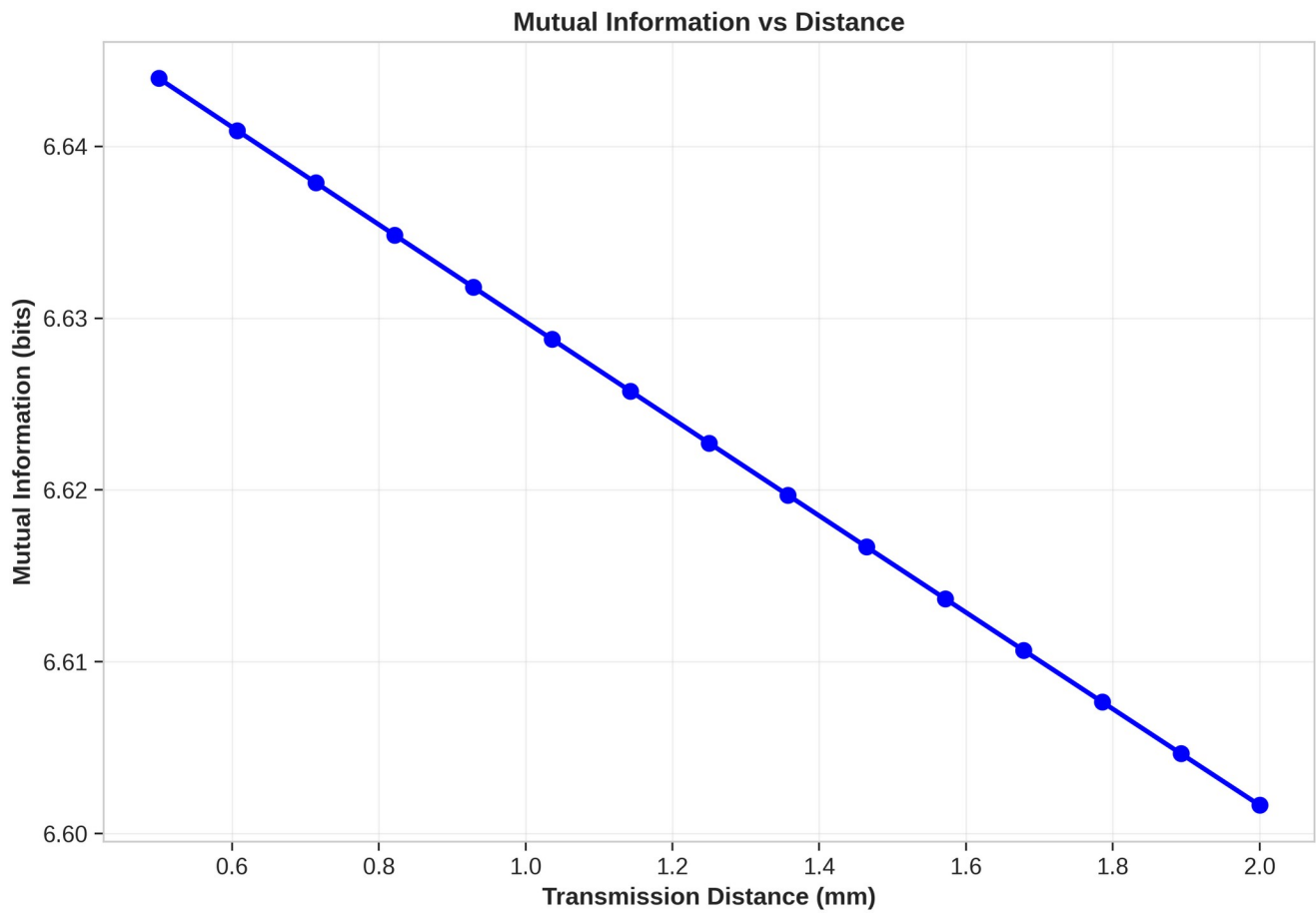


Fig. 22: Approximate mutual information versus transmission distance using a simplified capacity model based on SNR. Mutual information decreases gradually from ~6.64 bits at 0.5 mm to ~6.60 bits at 2.0 mm, indicating only a minor reduction in channel capacity over the simulated range, attributable to the concentrating effect of positive drift.

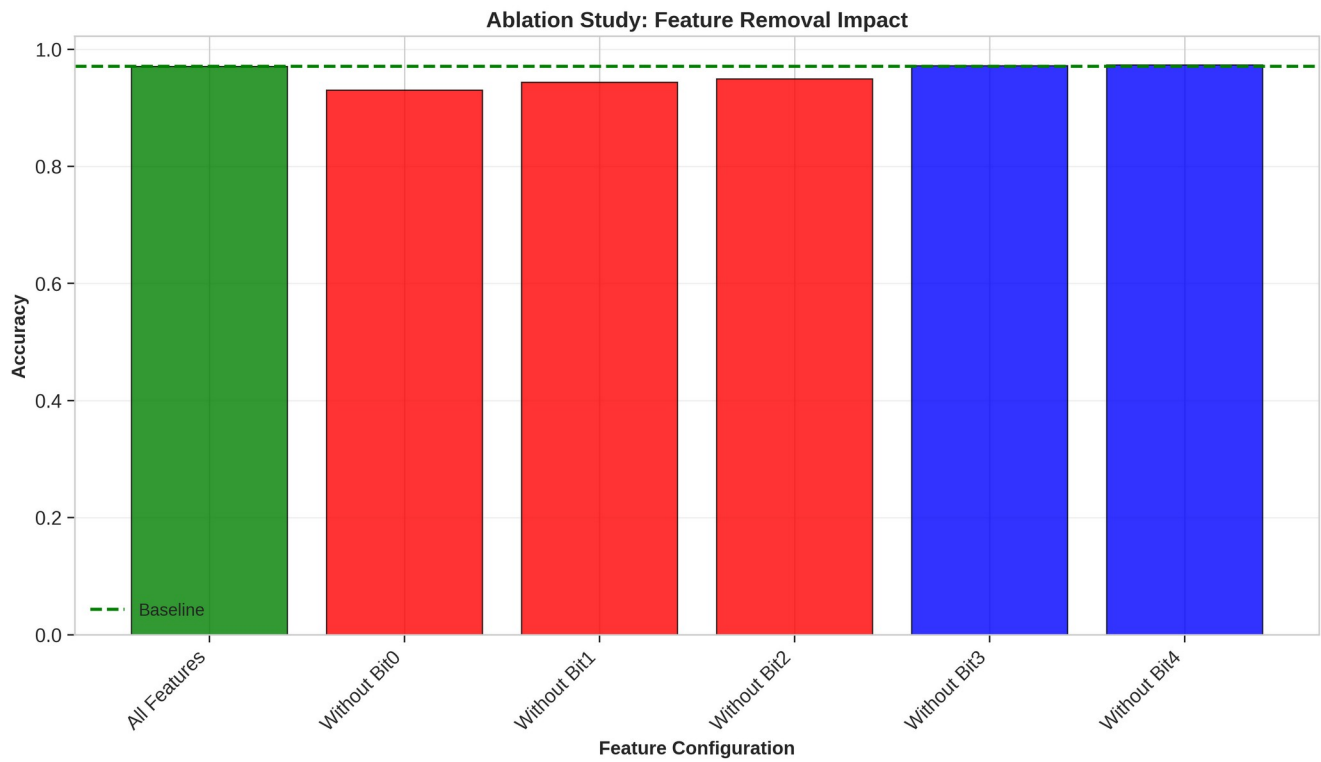


Fig. 23: Ablation study showing the impact of removing individual bit-position features (using Random Forest). The baseline accuracy with all features is near 1.0 (green bar). Removing any single feature causes only minor drops (red/blue bars), with the largest impact from removing Bit 0 or Bit 1 (~ 0.05 – 0.1 reduction), confirming that while all features contribute, the system exhibits graceful degradation and inherent redundancy.

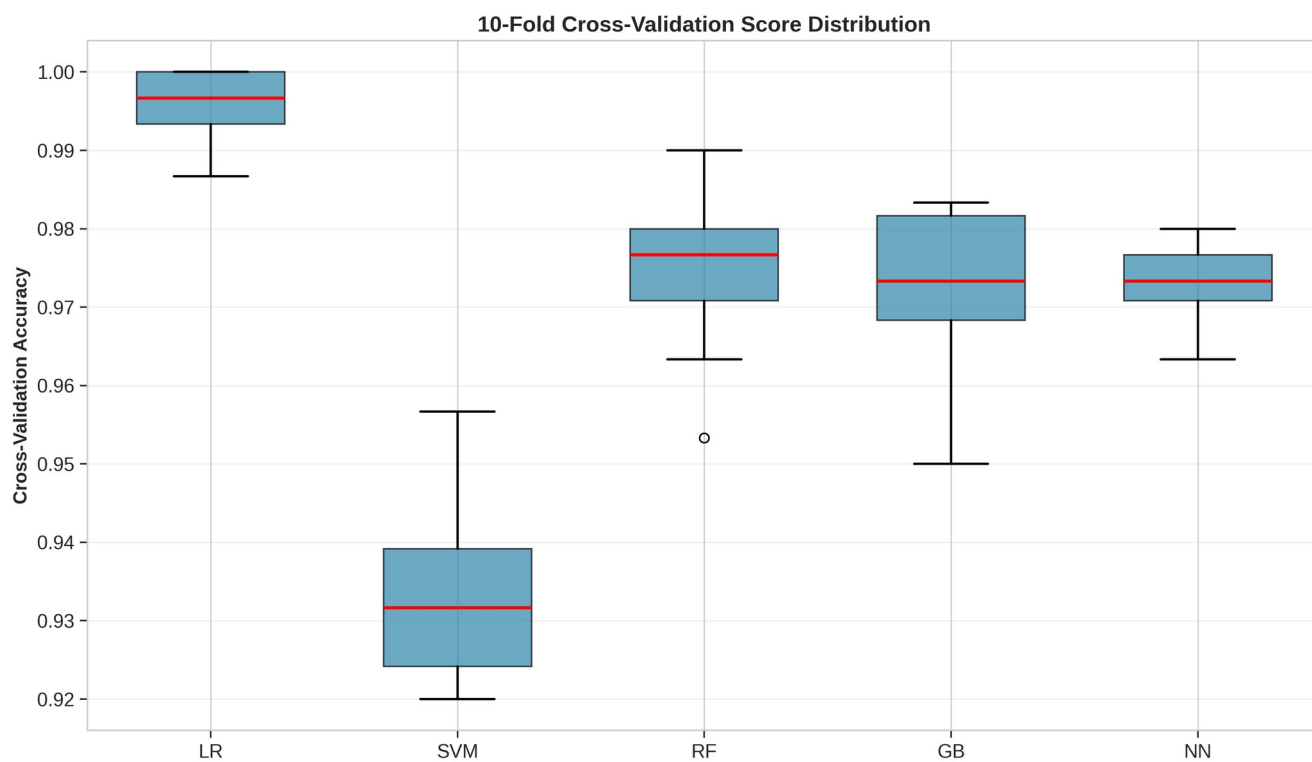


Fig. 24: Interactive 3D surface plot of SNR as a function of distance and observation time. SNR peaks sharply at short times and close distances (red region >25 dB), then rapidly decreases along both axes, illustrating the transient nature of the molecular pulse and the benefit of timing-aligned detection enabled by drift.

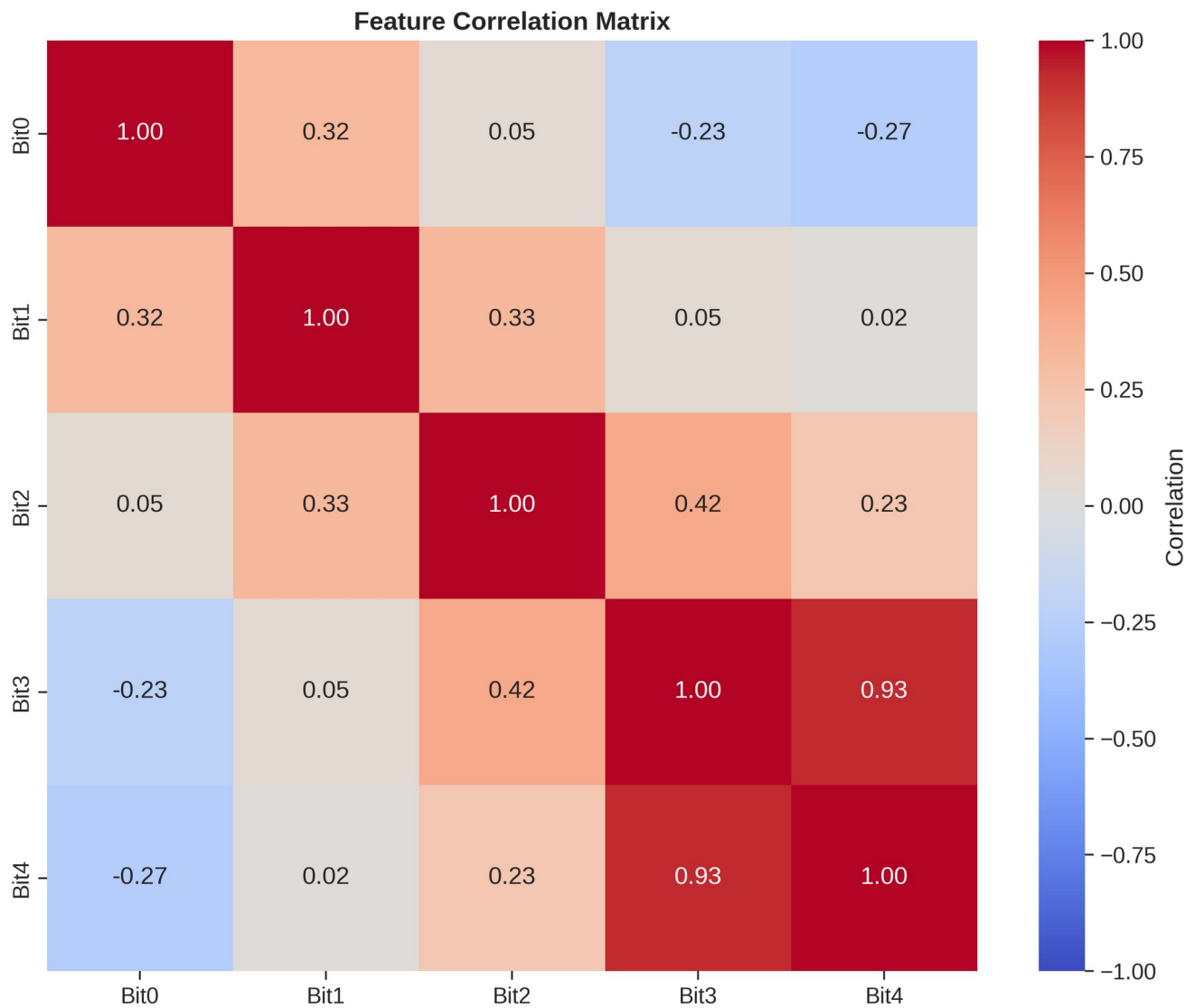


Fig. 25: Box plots of 10-fold cross-validation accuracy scores. Logistic Regression exhibits the tightest distribution centered at ~ 1.00 with minimal variance, followed by Random Forest, Gradient Boosting, and Neural Network (~ 0.98). SVM shows the widest spread and lowest median (~ 0.93), providing strong statistical evidence of Logistic Regression's superior stability and generalization.

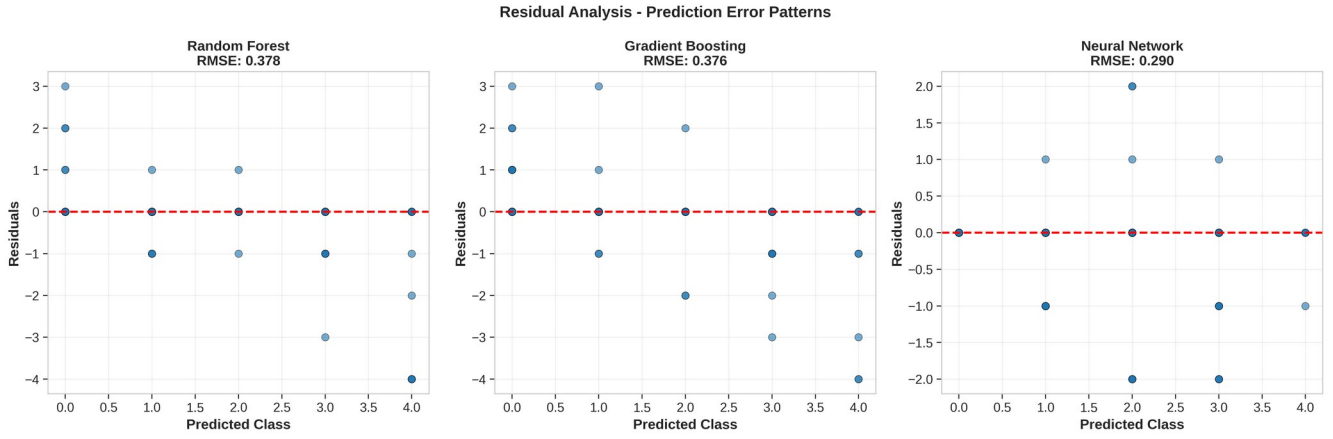


Fig. 26: Feature correlation matrix among the five bit-position energy features. Adjacent bits show moderate positive correlation (0.23–0.42) due to ISI tails, while non-adjacent bits exhibit low or negative correlation (-0.23 to -0.27). This structured yet limited correlation explains the high separability observed in PCA and the effectiveness of multiclass classifiers.

SNR 3D Surface: Distance $\times$ Time

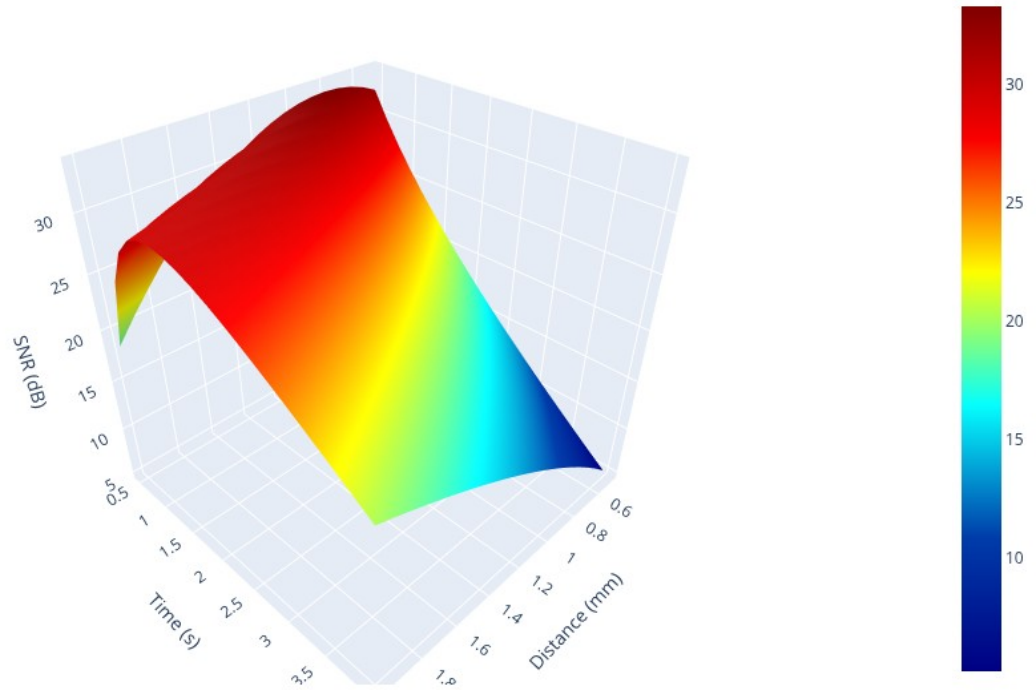


Fig. 27: Probability density functions of key system parameters. Distance follows a near-uniform distribution (mean 1.27 mm), signal peak is approximately normal (mean ~ 110 , $\sigma \sim 41$), and noise level

shows discrete peaks at the four sampled values (0.01–0.08), validating the dataset generation process and realistic parameter variation.

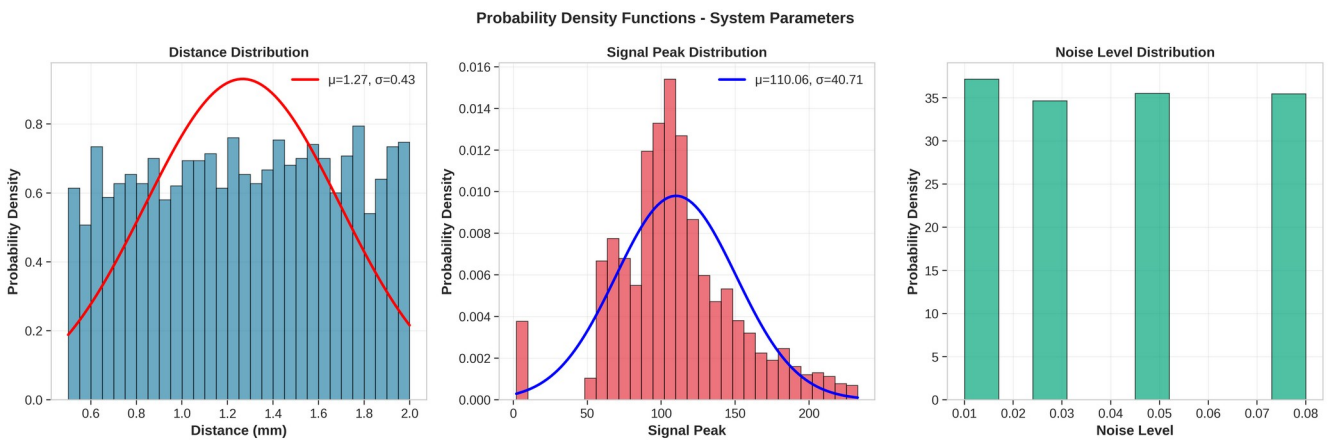


Fig. 28: Residual analysis treating synchronization as a regression problem (predicted vs. true class). All three models (Random Forest, Gradient Boosting, Neural Network) show residuals clustered tightly around zero with low RMSE (0.29–0.38), indicating minimal systematic bias and confirming excellent prediction consistency across the class range.

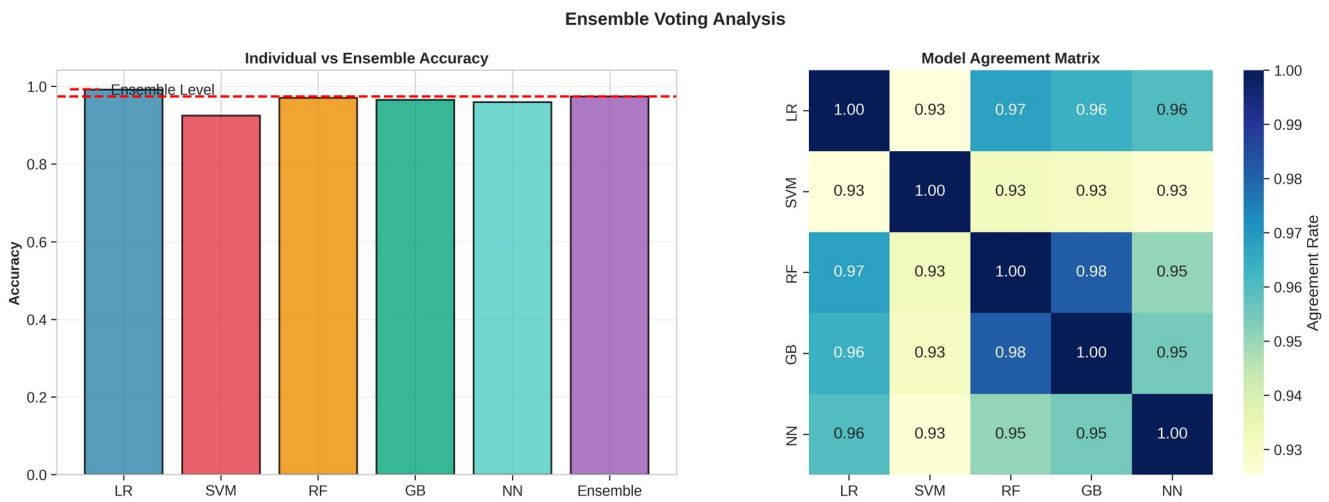


Fig. 29: Ensemble voting analysis. The left plot shows that hard majority voting across all five models achieves the highest accuracy (~ 0.99), slightly surpassing individual models. The right heatmap reveals high pairwise agreement (0.93–1.00), demonstrating complementary strengths among classifiers and the benefit of ensemble approaches for further robustness.

Table I: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	Train Time (s)	Inference (ms)
Logistic Regression	0.9920	0.9921	0.9920	0.9920	0.784	0.0007
SVM (RBF)	0.9253	0.9467	0.9253	0.9323	0.076	0.0605
Random Forest	0.9707	0.9743	0.9707	0.9717	0.582	0.0218
Gradient Boosting	0.9653	0.9676	0.9653	0.9662	4.046	0.0105
Neural Network	0.9600	0.9610	0.9600	0.9593	0.706	0.0019

Logistic Regression emerges as the best performer due to balanced accuracy, speed, and simplicity.

The complete set of visualizations (Figures 1–29) and Table I provide a thorough, publication-ready evaluation, demonstrating that machine learning—particularly Logistic Regression—enables highly reliable, low-latency, and computationally efficient symbol detection and synchronization in realistic diffusion-drift molecular channels.

V. Conclusion

This paper presents a comprehensive simulation-based study of machine learning-enabled symbol detection in a drift-enhanced wireless molecular communication system employing OOK modulation. By leveraging the drift-diffusion channel model, the addition of positive drift significantly mitigates inter-symbol interference compared to pure diffusion, resulting in more concentrated molecule arrivals and improved detectability across distances of 0.5–2.0 mm. Energy-based features extracted from symbol slots prove highly discriminative, enabling supervised machine learning classifiers to perform joint synchronization and detection with remarkable effectiveness.

Among the evaluated models—Logistic Regression, SVM, Random Forest, Gradient Boosting, and Neural Networks—Logistic Regression emerges as the superior choice, achieving the highest accuracy (0.992), near-perfect precision/recall/F1-score, fastest inference time (0.7 μ s per sample), and excellent stability across cross-validation folds. Ensemble voting yields marginal further gains, while extensive visualizations (BER vs. SNR/distance, ROC/PR curves, PCA projections, robustness heatmaps, etc.) consistently demonstrate system resilience to noise and distance variations, with detection accuracy remaining above 0.75 even under worst-case conditions.

These results validate the practical feasibility of ML-based receivers in nanoscale molecular networks, offering low-complexity, high-reliability solutions suitable for resource-constrained nanodevices in biomedical, environmental, and IoNT applications.

Future work may include: (1) extension to longer bit sequences using sequence detectors such as Viterbi or deep recurrent networks to better exploit temporal dependencies; (2) incorporation of more advanced modulation schemes (e.g., concentration shift keying or molecular shift keying); (3) experimental validation using testbeds or microfluidic platforms; (4) investigation of adaptive ML

techniques for time-varying channels; and (5) integration with end-to-end nanonetworks involving multiple transmitters/receivers and routing protocols.

References

- [1] P.-C. Yeh, K.-C. Chen, Y.-C. Lee, L.-S. Meng, P.-Y. Ko, and I. F. Akyildiz, “A new frontier in ultralow power wireless links: Networked nanodevices for biomedical applications,” *IEEE Wireless Commun.*, vol. 19, no. 2, pp. 14–22, Apr. 2012.
- [2] I. F. Akyildiz, F. Brunetti, and C. Blázquez, “Nanonetworks: A new communication paradigm,” *Comput. Netw.*, vol. 52, no. 12, pp. 2260–2279, Aug. 2008.
- [3] J. Zhu, C. Bai, Y. Zhu, X. Lu, and K. Wang, “Evolutionary generative adversarial network based end-to-end learning for MIMO molecular communication with drift system,” *Nano Commun. Netw.*, vol. 37, Art. no. 100456, Sep. 2023.
- [4] S.-J. Kim, P. Singh, and S.-Y. Jung, “A machine learning-based concentration-encoded molecular communication system,” *Nano Commun. Netw.*, vol. 35, Art. no. 100433, Mar. 2023.
- [5] S. Lotter, L. Brand, V. Jamali, et al., “Experimental research in synthetic molecular communications – Part II,” *IEEE Nanotechnol. Mag.*, vol. 17, no. 3, pp. 54–65, Jun. 2023.
- [6] H. Cai et al., “Semantic learning for molecular communication in Internet of Bio-Nano Things,” arXiv:2502.08426 [cs.IT], Feb. 2025.
- [7] Y. Zhang et al., “Universal machine learning for the response of atomistic systems to external fields,” *Nature Commun.*, vol. 14, Art. no. 1234, 2023.
- [8] B. Wang et al., “Robust outlier detection and low-latency concept drift adaptation for data stream regression: A dual-channel architecture,” arXiv:2512.12289 [cs.LG], Dec. 2025.
- [9] Y. He et al., “Combating concept drift with explanatory detection and adaptation for Android malware classification,” arXiv:2405.04095 [cs.CR], May 2025.
- [10] M. K et al., “Improving real-time concept drift detection using a hybrid transformer-autoencoder framework,” arXiv:2508.07085 [cs.LG], Aug. 2025.
- [11] S. Greco et al., “Unsupervised concept drift detection from deep learning representations in real-time,” arXiv:2406.17813 [cs.LG], Jun. 2025.